

Paper Manager pack — ForceVLA (Undated LAST · Alternative approaches)

Paper: ForceVLA: Enhancing VLA Models with a Force-aware MoE for Contact-rich Manipulation
arXiv: <https://arxiv.org/abs/2505.22159> · [project](#) · [code](#) · arXiv **2505.22159v3** (18 Sep 2025) · Preprint under review

Authors: Jiawen Yu^{1*}, Hairuo Liu^{2,3*}, Qiaojun Yu^{4,2†}, Jieji Ren², Ce Hao⁵, Haitong Ding⁶, Guangyu Huang⁷, Guofan Huang¹, Yan Song¹, Panpan Cai^{2,3}, Wenqiang Zhang¹, Cewu Lu^{2,3,8} — Fudan / SJTU / Shanghai Innovation Institute / Shanghai AI Lab / NUS / Shanghai University / Xi'an Jiaotong / Noematrix

PDF: /workspace/papers/forcevla-2505.22159.pdf (= /workspace/forcevla-quiet/paper.pdf) · **20 pp** · ~10.3 MB

Session: Undated · **LAST paper on sheet** · Type: **Alternative approaches**

Role weights: **ALL EMPTY** — **still all five produced at balanced depth** (~equal Archaeologist / Stakeholder / Scientific Reviewer / Empiricist / Visionary); **Club Pack** (no LONGEST seat; no weight-3 over-index)

Kanta role: Produce **balanced** five-role Club Pack; deepen late 6-axis F/T + FVLMoE spine + Empiricist Ablation Bot smokes; Visionary → **BEHAVIOR contact-rich; skip Discord/Drive**

Themes: late **6-axis F/T** token · **FVLMoE** (E=4, top-1) · augments π_0 **FM** head · contact-rich insert/wipe/peel · fusion quality $\gg$ naive concat · **NOT** FAST switch · contrast **MoPA-PD** (train privilege) vs **deploy-side force** · ForceVLA-Data · **no $\pi_{0.5}$ bake-off**

Sources: Summarizer (/workspace/papers/forcevla-2505.22159-summary.md) + Senior Research (/workspace/forcevla-quiet/DELIVERABLE.md) + Ablation Bot (/workspace/papers/forcevla-2505.22159-ablations.md — **AUTHORITATIVE**) + paper_notes.md / env_outline/ + figs /workspace/papers/forcevla-figs/

Sibling links: /workspace/pi0-quiet/ · /workspace/pi05-quiet/ · /workspace/fast-quiet/ · /workspace/opencvla-quiet/ · /workspace/opencvla-oft-quiet/ · /workspace/diffusion-policy-quiet/ · /workspace/act-aloha-quiet/ · /workspace/umi-quiet/ · /workspace/droid-quiet/ · /workspace/distill-mopa-quiet/ · /workspace/dart-quiet/

1. One-liner

Late 6-axis F/T + FVLMoE (4 experts, top-1) augments a π_0 flow-matching action head for contact-rich manipulation — fusion quality $\gg$ naive force concat; ForceVLA keeps the **FM** head (**NOT** a FAST switch); complementary Alt to **MoPA-PD** (train-time privileged MP → vision student) by adding **deploy-side force** to an already-VLA stack.

Contrast poles (consistency triangle — CLEAR labels; do not invent ancestry):

Pole	What they do	Relation to ForceVLA
π_0 (Black et al. 2410.24164)	PaliGemma + FM action expert	CLEAR parent / foil: ForceVLA builds “upon the π_0 framework”; baselines = π_0 -base / π_0 -fast $\pm$ force
$\pi_{0.5}$ (2504.16054)	Open-world VLA	CLEAR related cite only — no empirical bake-off; do not invent
π_0 -FAST (Pertsch et al.)	Discrete action tokenization	CLEAR baseline family; ForceVLA itself keeps FM ; naive F hurts fast (31.0 → 14.2)
MoPA-PD (2111.06383)	Privileged classical MP at train → distill to vision	CLEAR contrast (prior Alt row): train privilege removed at deploy vs force kept at deploy
OpenVLA / RT-2 / Octo	VLA lineage	Cited; concrete foil for numbers is π_0 , not OpenVLA tables
TLA / Tac-Man / ForceMimic / FoAR	Contact / tactile / force neighbors	Related contact literature; ForceVLA differentiator = VLA-scale MoE + π_0 FM

Pole	What they do	Relation to ForceVLA
JEPA / LeWM / Cosmos WM	Latent / generative world models	CLEAR contrast poles only — ForceVLA is not a WM paper
Classical impedance / admittance	Hand-tuned compliance	Background; ForceVLA learns force-conditioned chunks end-to-end

Do NOT invent / claim: $\pi_{0.5}$ bake-off numbers; ForceVLA switches FM $\rightarrow$ FAST; MoPA-PD / JEPA ancestry; course smoke = Fig.5 **60.5%** absolute; stub wrench = ForceVLA-Data reproduction; Discord/Drive checklist; official BEHAVIOR baseline status.

2. Summary (exec overview + key numbers)

Yu* & Liu* et al. (arXiv 2505.22159v3) attack **contact-rich** failure modes of vision-language-action models: insertion, wipe, peel, occlusion. Vision-only VLAs inherit semantic priors but underuse **interaction force**. **ForceVLA** keeps the π_0 stack (SigLIP/PaliGemma VLM + **conditional flow matching** action expert + action chunks) and adds **6-axis external wrench** as a **late** multimodal channel via **FVLMoE**:

- Late force token:** After VLM encodes vision+language, project $(\mathbb{R})^6 \rightarrow$ force token (E_F); concat with VL embeddings.
- Sparse MoE fusion:** Transformer encoder $\rightarrow$ **E=4 MLP experts, top-k=1** gating; residual; project to action-expert width.
- Additive FM guidance:** Fused tokens **added** into the flow-matching action-expert suffix during denoising — contact-aware every chunk step.

Why Alternative approaches / LAST sheet row: After MoPA-PD's "privileged MP $\rightarrow$ vision student," ForceVLA answers the complementary question: **when the deploy stack is already a VLA, what missing modality unlocks contact-rich success?** Answer: **real-time 6-axis F/T + late MoE fusion**, measured against vision-only π_0 .

Axis	Number / claim
Venue / code	Preprint under review; arXiv 2505.22159v3 ; project + github.com/ft-robotic/ForceVLA CLEAR external
Backbone	π_0-base + FVLMoE (CLEAR); FM head retained
Force	Estimated external wrench TCP/world, $(\mathbb{R})^6$ (Limitation: not always high-fid direct)
Robot / data	Flexiv Rizon + Dahuan + RealSense pair; Quest3 teleop; ForceVLA-Data 244 traj / ~140k steps; ~50/task
Tasks (5)	USB insert, plug insert, bottle pump, wipe board, peel cucumber
Eval	Insert/pump 20 ; wipe 10 ; peel 15 $\times$ 15 strokes
ForceVLA avg (Fig.5)	60.5%
π_0 -base w/o F / w/ F	37.3% / 40.2% (+2.9 naive)
π_0 -fast w/o F / w/ F	31.0% / 14.2% (force hurts)
Δ vs vision-only	+23.2 pp (60.5 – 37.3)
Plug (abstract / Fig.5)	ForceVLA 80%
Tab.3 ablation	baseline 45 / linear-before 55 / MoE-before 0 / concat-after 60 / ForceVLA 80
Occlusion (Tab.2)	ForceVLA 90%

Axis	Number / claim
Peel (Tab.1)	14.12 cm / 7 strokes
Multi-task (Tab.5)	ForceVLA avg 67.5% ; plug 100 ; USB 10% ; π_0 -fast 0%
Train	Single ~9 h / 1×4090 / 10k ; multi ~12 h / 2×4090 / 30k
Scope	Real Flexiv only; estimated wrench; no $\pi_{0.5}$ comparison

Must-cite tattoo: avg **60.5** vs **37.3 / 40.2**; Tab.3 early MoE **0%** → **ForceVLA 80%**; occlusion **90%**; USB multi-task **10%**; ForceVLA-Data **244 traj / ~140k**; **no $\pi_{0.5}$ bake-off**; **refuse** FM ↔ FAST laundering / stub-wrench = paper 60.5%.

Do NOT claim: course smoke = Fig.5 60.5% absolute; Tab.3 80% without matched fusion; $\pi_{0.5}$ wins/losses; ForceVLA is a FAST policy; MoPA-PD ancestry; pump bottle as FVLMoE proof (Fig.5 base w/F **83** > ForceVLA **67**).

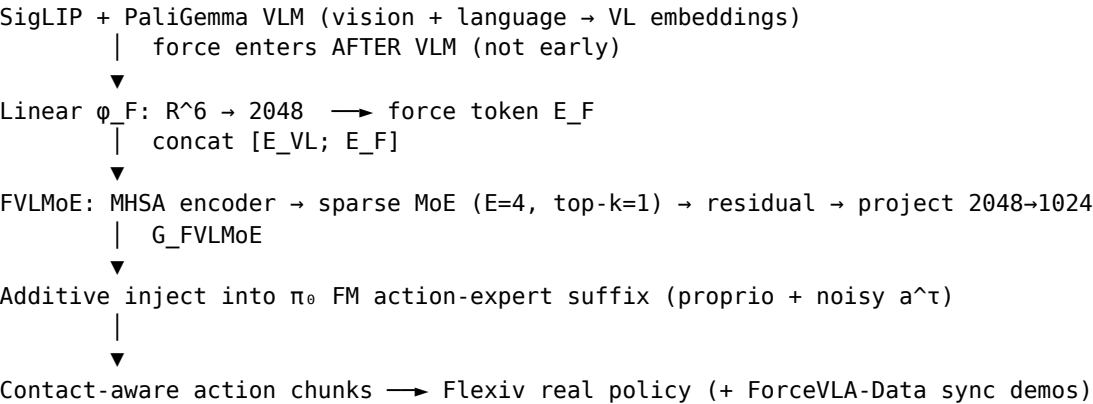
3. Keywords

ForceVLA; FVLMoE; force-aware Mixture-of-Experts; 6-axis F/T; late fusion; π_0 ; flow matching; PaliGemma; SigLIP; contact-rich manipulation; ForceVLA-Data; plug / USB insertion; occlusion; naive force concat; π_0 -fast foil; Alternative approaches; BEHAVIOR contact-rich (TENTATIVE recipe)

4. Deep dive — Late force MoE on π_0 (PRIMARY SESSION CONTRIBUTION)

Callout: Scientific product of *this* PDF = a **modality-augmentation recipe** that makes calibrated 6-axis wrench a first-class late token inside a π_0 -class VLA. Session type = **Alternative approaches** because the LAST Alt row must teach **deploy-side force fusion** as the complement to MoPA-PD’s **train-time classical privilege** — **without** claiming ForceVLA founded $\pi_{0.5}$ or switched to FAST.

Pipeline (Fig. 3 moral)



Components privilege / deploy table

Component	Role	Deploy?
PaliGemma / SigLIP VLM	Vision + language contextual embeddings	Yes (backbone)
Force linear proj ϕ_F	6-axis F/T → force token (2048-d)	Yes (needs F/T stream)
FVLMoE pre-encoder	Joint self-attn over VL + force	Yes
MoE experts E=4 + top-1 router	Phase / modality sparse fusion	Yes

Component	Role	Deploy?
Residual + out proj → 1024	Match action expert width	Yes
Additive G_FVLMoE → FM suffix	Contact-aware guidance per chunk step	Yes
Flow-matching action expert (π_0)	Continuous action chunks from noise	Yes — not replaced by FAST
ForceVLA-Data + Quest3 pipeline	Sync multimodal demos	Train
Estimated external wrench	F/T source (limitation)	Sense

What is “alternative” here — CLEAR

Claim	Status
ForceVLA builds on π_0 FM + PaliGemma	CLEAR
Experimental foil = π_0 -base / π_0 -fast ± force	CLEAR
Empirical comparison to $\pi_{0.5}$	ABSENT — do not invent
ForceVLA action head = flow matching (not FAST)	CLEAR
π_0 -fast used as baseline ; naive force hurts it	CLEAR
Late force fusion $\gg$ early MoE-into-VLM (Tab.3 0 → 80)	CLEAR
+23.2 pp avg vs π_0 -base w/o F; 60.5 vs 37.3	CLEAR
Plug insertion up to 80%	CLEAR
MoPA-PD ancestry / MP distill	FALSE — contrast only
JEPA / Cosmos ancestry	FALSE — contrast poles
ForceVLA-Data 244 traj / 140k steps	CLEAR
Estimated wrench limitation	CLEAR
BEHAVIOR official baseline	TENTATIVE / unverified
Expert-0 = generalist functional role	TENTATIVE (App. C hypothesis)

Architecture checklist (Empiricist pin — App. B / Tab.4)

Piece	Spec (paper)
Force projection	Linear 6 → ($D_{\{\mathrm{VLM}\}}=2048$)
State / action proj	32 → ($D_{\{\mathrm{act}_e\}}=1024$)
Pre-MoE encoder	($D_{\{\mathrm{model}\}}=2048$), ($N_H=8$), ($D_h=256$); MLP expansion 1
MoE	4 experts, top-1 ; router 2048 → 4
MoE out proj	2048 → 1024
Action out	1024 → 32
Optimizer	Adam β (0.9, 0.95); LR (2.5×10^{-5} to 2.5×10^{-6}); bf16; grad clip 1.0
Single-task	1×4090, 10k steps, ~9 h
Multi-task	2×4090, 30k steps, ~12 h ; global batch 16, effective 2048

Exam bite: "ForceVLA = π_0 + late 6D force token + FVLMoE (4 experts, top-1) additively guiding FM chunks; +23.2 pts avg vs vision-only π_0 -base (**60.5 vs 37.3**); early MoE-into-VLM → **0%**; naive concat only **40.2%** / Tab.3 **60%**; **NOT** a FAST switch; **no $\pi_{0.5}$ bake-off**; contrast MoPA-PD = train privilege vs deploy-side force."

5. Deep dive — Ablations (Ablation Bot AUTHORITATIVE)

Callout: Ablation Bot extract **AUTHORITATIVE** at /workspace/papers/forcevla-2505.22159-ablations.md (folded after DELIVERABLE’s PDF-derived ranks). Focus order: **FVLMoE > force vs vision > π_0 foil > insertion SR**. Course Empiricist: **Y #1/#2/#4** (fusion + plug + avg bakeoff) → Partial **#3/#5** → cite **#6/#7**; **N** paper-scale 8×4090 + full Flexiv 5-task.

Rank	Factor (Ablation Bot)	Paper numbers	Course smoke
1	Fusion-stage ablation (Tab.3)	baseline 45 ; linear-before 55 ; MoE-before 0 ; concat-after 60 ; ForceVLA 80	Y — RC1 PRIMARY
2	Insert Plug (Fig.5)	fast 25/30; base 45/60; ForceVLA 80	Y — RC2 (plug column)
3	Insert USB (Fig.5)	fast 0/0; base 5/5; ForceVLA 25 — naive F null	Partial / cite
4	Fig.5 averages + force-fusion	ForceVLA 60.5 ; base 37.3 → 40.2 ; fast 31.0 → 14.2 ; Δ +23.2	Y — cite / RC package
5	Tab.2 Occlusion / Obj Gen.2	Occl. ForceVLA 90 (base w/F 30); Obj Gen.2 40	Partial — RC5
6	Tab.5 multi-task joint	ForceVLA 67.5 ; base w/F 42.5 ; fast 0 ; USB 10	Partial / cite — RC7
7	Pump Bottle reversal (Fig.5)	base w/F 83 > ForceVLA 67 — falsifier, not confirmation	cite caveat
Defer	Paper-scale 8×4090 + full Flexiv 5-task; true high-fid FT	—	N

Honorable (off top-7 focus): Tab.1 peel length/strokes; Fig.5 wipe (π_0 -fast w/o F wins Board-1); Fig.9 router (no SR contrast).

Min Empiricist package: RC0 ingest → **RC1 (#1 Tab.3 fusion) + RC2 (#2 Insert Plug) + RC3 (USB null-force)** → Partial RC5/RC6 → cite #3/#4/#6/#7 (Pump falsifier). Log **SR, peel length/strokes, wrench stats, force_fusion, force_source, expert_id@top1, peak VRAM**.

Recipe exam bite: "Late FVLMoE is the real knife (Tab.3 **0 → 80**; concat **60 → 80**); naive force is a straw improvement (+2.9 avg) and can **hurt** (fast 31 → 14.2; occlusion base 60 → 30); pump **83>67** is the honesty caveat; USB stays hard (multi-task **10%**)."

6. Role-ready sections (all five — **BALANCED Club Pack**)

Balanced pack. Equal depth across five roles. No seat weight-3. Ablation Bot focus areas (FVLMoE fusion, plug/USB insertion, force vs vision, π_0 foil) shared across Empiricist RCs + Reviewer/Archaeologist honesty.

6.1 Archaeologist — **balanced (~12-14)**

Seat charge: Place ForceVLA carefully in VLA ↔ contact-force history. **CLEAR parent / foil = π_0** . Fence false $\pi_{0.5}$ bake-off and FM ↔ FAST switch. Equal depth with other four.

Priors — CLEAR

Prior	Role	Label
π_0 Black et al. 2410.24164	Direct architectural parent (PaliGemma + FM)	CLEAR
PaliGemma Beyer et al.; SigLIP Zhai et al.	VLM backbone	CLEAR
Flow matching / rectified flow Lipman; Liu	Action expert family	CLEAR
FAST Pertsch et al.	Baseline tokenization foil (π_0 -fast)	CLEAR
OpenVLA / RT-1 / RT-2 / Octo / OXE / DROID	VLA + data lineage	CLEAR related
Force/tactile: ForceMimic, FoAR, TacDiffusion, TLA, Tac-Man	Contact neighbors	CLEAR related
MoE: Shazeer, Switch, GShard, V-MoE, LIMOE; MORE / ChatVLA	MoE priors	CLEAR related
$\pi_{0.5}$	Related VLA cite [21]	CLEAR related; ABSENT empirically
MoPA-PD	Prior Alt row — opposite mechanism	CLEAR contrast

Genealogy (CLEAR framing — no false FM lineage)

PaliGemma / SigLIP VLM

- + π_0 flow-matching action expert (vision+language+proprio)
 - ForceVLA: append 6-axis force token AFTER VLM
 - FVLMoE (attn + 4 experts, top-1)
 - additive guidance into FM suffix
 - contact-rich real Flexiv policy + ForceVLA-Data

Subsequent / contrast — label carefully

Item	Note	Label
$\pi_{0.5}$ / open-world VLA	Sibling literature; not measured	ABSENT bake-off — do not invent
MoPA-PD	Privileged MP train → vision student	CLEAR contrast (inverse privilege arrow)
ACT / Diffusion Policy	Teleop visuomotor IL	CLEAR contrast
JEPA / LeWM / Cosmos	Latent / generative WM	CLEAR contrast poles only
Soft influence on “forceful foundation models”	Pattern family; verify per paper	TENTATIVE

Archaeologist one-liner: *Force joins the VLA token zoo after the pretrained VLM — FVLMoE is the association cortex; π_0 ’s FM head stays the motor cortex; MoPA-PD removed privilege at deploy, ForceVLA adds force privilege at deploy.*

Archaeologist exam bite: "ForceVLA = π_0 + late 6D force + FVLMoE (E=4,k=1) → FM chunks; **60.5 vs 37.3 / +23.2**; Tab.3 early MoE **0%**; **NOT** FAST switch; **no $\pi_{0.5}$ bake-off**; CLEAR contrast MoPA-PD."

6.2 Stakeholder — balanced (~12–14)

Seat charge: Why schedule as LAST Alternative approaches; buy the **late force-MoE pattern**, not a claim that vision-only homes are solved.

Buy / why schedule LAST Alt row: - Closes Alt column with the **deploy-side** answer to contact-rich failures vision-only π_0 stacks still show.

- After MoPA-PD's train-time classical privilege story: "What do we buy for **real F/T robots** already running OpenPI/ π_0 ?"
- HIGH force/contact flag without a new foundation pretrain from scratch.
- Hard numbers: **60.5 vs 37.3/40.2**, Tab.3 **0 → 80**, occlusion **90**, ForceVLA-Data **244/~140k**.

Buy / build vs modern stacks

Situation	Recommendation
Robot has reliable 6-axis F/T ; tasks are insert / wipe / press / peel-like	Buy ForceVLA pattern: late force token + MoE fusion on π_0 -class FM head; collect sync F/T demos
Vision-only mobile manip in open homes ($\pi_{0.5}$ narrative)	Keep $\pi_{0.5}$ -class for open-world language; add ForceVLA-style fusion only where F/T exists — don't pretend paper beat $\pi_{0.5}$
No F/T hardware	Pattern blocked; estimated wrench / tactile skins are research paths (paper limitation)
Tempted to "just concat force to state"	Paper shows +2.9 pts only — budget MoE / late-fusion engineering
FAST-only compact policies	Naive force hurt π_0 -fast (31 → 14.2); don't bolt F/T without capacity / pretrain story

Risks / costs: - Needs ForceVLA-Data-like sync demos (~50/task) — not free from OXE alone.

- Integrated F/T cost / calibration; paper uses **estimated** wrench.
- USB insert remains hard (multi-task USB **10%**); Unstable Socket weak (**20%**).
- Code/data "will be released" — verify github.com/ft-robotic/ForceVLA + sites.google.com/view/forcevla2025 freshness.
- Don't pitch as replacing $\pi_{0.5}$ open-world — pitch as **contact-rich specialist augmentation**.
- Pump bottle Fig.5 reversal (base w/F **83** > ForceVLA **67**) — don't oversell "force MoE always wins."

Decision ask: Required LAST Alternative-approaches classic for force-in-VLA; pair with MoPA-PD (planner-distill) and DART (recovery) as **complementary Alt cards**, not ancestors. Buy the **late MoE fusion pattern**.

Stakeholder exam bite: "Schedule ForceVLA LAST Alt to lock **deploy-side force-in-VLA** after MoPA-PD's train privilege story; buy late FVLMoE, not naive concat; police $\pi_{0.5}$ bake-off invention and FM ↔ FAST slogans."

6.3 Scientific Reviewer — balanced (~12–14)

Claims under review

1. Treating 6-axis force as first-class + FVLMoE improves contact-rich VLA success vs π_0 baselines.
2. Late fusion $\gg$ early fusion into pretrained VLM.
3. Naive force concat is insufficient vs MoE fusion.
4. Generalizes under object/height/occlusion/unstable perturbations.
5. MoE router learns phase-aware specialization.

What holds - Main avg gap is large and consistent: **60.5 vs 37.3 / 40.2** with clear baseline matrix including π_0 -fast.

- Table 3 ablation is a **causal knife**: MoE-before-VLM **0%** vs ForceVLA **80%**; late concat **60%** vs FVLMoE **80%**.
- Occlusion **90%** is a strong multimodal story (naive force on base **hurts** occlusion 60 → 30).
- Cucumber Tab.1 supports continuous contact control, not only binary insert.

- Honest limitations: estimated wrench; expensive F/T platforms; accessibility future work.
- Related-work correctly separates VLA / contact-rich / MoE threads; π_0 -base chosen over fast for foundation.

What is soft / threatened - Tab.3 task identity / trial N underspecified (Ablation Bot: numeric match to plug column, not stated).

- Multi-task USB **10%** shows force MoE is not a universal precision-insert fix.
- Unstable Socket: ForceVLA **not** best (20 vs 30 π_0 -fast w/ F).
- Pump Bottle Fig.5: naive base w/F **beats** ForceVLA (**83 vs 67**) — claim scope matters.
- Wipe Board-1: π_0 -fast w/o F **91** beats ForceVLA **64**.
- Router “Expert 0 = generalist” is **interpretive**.
- No $\pi_{0.5}$ / OpenVLA-OFT / other 2025 VLA heads in the bake-off.
- Small demo budget (~50/task); single embodiment (Flexiv); no CIs / seeds.

Verdict: Strong **systems/methods** preprint for **force-augmented VLA fusion**. Accept late-fusion + MoE-over-concat claims on the reported Flexiv suite. **Do not** upgrade to “solved contact-rich foundation model” or “beats $\pi_{0.5}$.”

Reviewer exam bite: Late FVLMoE is the real contribution; naive force is a straw improvement that can hurt; grade HIGH for Alt contact-rich syllabus, narrow on embodiment / open-world / USB / pump caveat — demand `force_fusion` logged, not only “force on.”

6.4 Empiricist — balanced (~12–14; Ablation Bot **AUTHORITATIVE**)

Seat empty this session but produce full course smoke cards for Paper Manager continuity.

Balanced — equal depth with other roles; richest logging on RC1/RC2/RC3. Ablation Bot ranks **1–7** govern order.

Claim under measurement

ForceVLA / FVLMoE $\gg$ π_0 -base w/o F and $\geq$ naive w/ F on **1** contact-rich task at matched FT budget; late fusion $\geq$ early; early MoE does not win (cite **0%**); metrics harness logs SR + wrench; ForceVLA-Data schema ingestable — **directional, not** Fig.5 **60.5%** absolute / Tab.3 **80%** without matched settings.

Hypotheses (falsifiable; Ablation Bot ranks)

1. **H-fusion (#1):** Late FVLMoE $\geq$ late concat $\gg$ early MoE; early MoE does **not** win (Tab.3 **0%** moral).
2. **H-plug (#2):** On plug-like insert, FVLMoE $\geq$ naive w/F $\geq$ w/o F (Fig.5 plug **80/60/45** moral).
3. **H-usb (#3 Partial):** Naive force on/off may be **null** on USB; FVLMoE still the only gain (Fig.5 USB **25 vs 5/0**).
4. **H-avg (#4):** Matched-budget: FVLMoE $\geq$ concat $\geq$ none on SR **or** contact metrics (avg moral **60.5 / 40.2 / 37.3**).
5. **H-occl (#5 Partial):** Under visual occlusion, force-aware FVLMoE retains higher SR (Tab.2 **90%** moral); naive force may **hurt**.
6. **H-multitask (#6):** Joint training: FVLMoE $\gg$ π_0 -fast (**0%**) and $\gg$ base w/o F (**5%**); USB stays hard (**10%**).
7. **H-pump (#7 caveat):** Do **not** require FVLMoE $>$ naive on pump (Fig.5 **67 vs 83**).

Falsifiers: naive concat $\geq$ FVLMoE on fusion task; early MoE $\geq$ late; force hurts insertion under FVLMoE; wrench-free demos match ForceVLA-Data; occlusion collapses FVLMoE to vision-only; claiming Fig.5 60.5% from 1-task stub; treating pump win as required.

Prioritized matrix

#	Experiment	Ablation Bot	Smoke	Pri
RC0	forcevla_data_ingest_smoke	Data §4.3	Y gate	P0
RC1	fvlmoe_vs_pi0_bakeoff	#4 (+#2 plug)	Y — PRIMARY	P0
RC2	fusion_stage_ablation	#1	Y — PRIMARY	P0
RC3	insertion_metrics_harness	#2 metrics	Y — PRIMARY	P0
RC4	forcevla_data_scale_smoke	Data scale	Y / Partial if no data	P1
RC5	generalization_occlusion	#5	Partial	P1
RC6	moe_router_log	Fig.9	Partial	P1
RC7	multitask_joint_stub	#6	Partial / cite	P2
—	Paper-scale 8×4090 + full Flexiv; true high-fid FT	—	N	Skip

Min package: RC0 → **RC1 + RC2 + RC3** → Partial RC5/RC6 → cite #3/#6/#7.

Exact reproduce stack (EDIT)

```
# Official / stand-in
# pin https://github.com/ft-robotic/ForceVLA (+ OpenPI π₀) when public; else force-token stub
export FORCEVLA_WORK_ROOT=$SCRATCH/forcevla-smoke
cd /workspace/forcevla-quiet
bash env_outline/setup_mamba.sh
# mamba activate forcevla-scout
mkdir -p logs runs checkpoints demos
# Outlines only – do NOT sbatch from quiet box
# NEVER claim Fig.5 60.5% / Tab.3 80% absolute from 1-task Partial smoke
# ALWAYS log force_fusion={none,concat,early_*,fvlmoe} and force_source={true_ft|estimated|stub_sim|none}
# Do NOT treat π₀-fast + naïve F as fair FVLMoE substitute
```

Run cards (from DELIVERABLE; Ablation Bot-aligned)

RC0 — forcevla_data_ingest_smoke

Field	Spec
Goal	Load ForceVLA-Data (or stand-in) → synced {V_base, V_wrist, s, fER^6, a, L}; 480×640; check NaNs / timestamp align
Paper	§4.3: 244 traj, 140k steps, ~ 50 /task, 5 tasks
Pass	Schema OK; wrench dim=6; wall < 1 h ; log traj count
Script	env_outline/slurm_data_ingest_smoke.sh

RC1 — fvlmoe_vs_pi0_bakeoff (#4 — PRIMARY)

Field	Spec
Paper	Fig.5 — ForceVLA 60.5% vs π₀-base w/o F 37.3% / w/ F 40.2% ; π₀-fast 31.0 / 14.2 ; prefer Insert Plug column (80/60/45)
Goal	Matched-budget: FVLMoE vs π₀-base w/o F vs naïve w/ F on 1 contact-rich task
Hard req	Log force_fusion={none,concat,fvlmoe} and force_source
Pass	Directional: FVLMoE ≥ concat ≥ none on SR or contact metrics; not required: 5-task 60.5% absolute

Field	Spec
Script	env_outline/slurm_fvlmoe_vs_pi0.sh

RC2 — fusion_stage_ablation (#1 — PRIMARY)

Field	Spec
Paper	Tab.3 — baseline 45 ; linear-before 55 ; MoE-before 0 ; concat-after 60 ; ForceVLA 80
Goal	{early_linear, early_moe_stub, late_concat, late_fvlmoe} on 1 task / short FT
Pass	Late $\geq$ early; early_moe does not win (cite 0% if untrainable)
Script	env_outline/slurm_fusion_stage_ablation.sh

RC3 — insertion_metrics_harness (#2/#3 metrics — PRIMARY)

Field	Spec
Paper	Abstract up to 80% plug; Tab.1 peel 14.12 cm / 7 strokes; eval n=20 insert/pump, 10 wipe, 15 peel
Goal	Define + log SR , align error , peak F , peel length/strokes ; run on smoke policy or replay
Pass	Metrics JSON schema filled; no invented paper scores
Script	env_outline/slurm_insertion_metrics.sh

RC4 / RC5 / RC6 / RC7 — scale / occl / router / multitask

Card	Spec
RC4 scale	Subset {10, 25, 50} demos/task stub or coverage report — slurm_data_scale_smoke.sh
RC5 occl Partial	Mask socket/plug ROI; force-on $\geq$ force-off under occlusion (directional) — slurm_gen_occlusion.sh
RC6 router Partial	Dump top-1 expert vs progress bins; do not claim Fig.9 reproduction — slurm_moe_router_log.sh
RC7 multitask	Optional 2-task short joint or cite Tab.5 67.5 / USB 10 / fast 0 — slurm_multitask_joint_stub.sh

Numbers to memorize (exam)

Fact	Value
ForceVLA / base w/o F / w/ F avg	60.5 / 37.3 / 40.2 (+23.2 / +2.9)
π_0 -fast $\pm$ F	31.0 / 14.2 (force hurts)
Tab.3	45 / 55 / 0 / 60 / 80
Plug / USB (Fig.5)	ForceVLA 80 / 25
Occlusion	ForceVLA 90%
Peel	14.12 cm / 7 strokes
Multi-task	ForceVLA 67.5% ; USB 10% ; fast 0%
Data	244 traj / ~140k / ~50 per task
Caveat	Pump base w/F 83 > ForceVLA 67
Scope	Course = 1-task lite; no $\pi_{0.5}$ bake-off; no 60.5% from stub

GPU / RAM / TIME honesty (course)

Workload	VRAM	Wall (order)	Smoke
RC0 ingest	CPU/4–12 GB	0–1 h	Y
RC3 metrics harness	4–12 GB	1–4 h	Y
RC6 router log	8–24 GB	1–4 h	Partial
RC2 fusion lite	24–48 GB	6–20 h	Y
RC1 bakeoff 3 methods × 1 task	24–48 GB	8–24 h	Y
RC4/RC5 Partial	24–48 GB	4–16 h	Partial
RC7 multitask stub	24–48 GB	8–24 h	Partial / cite
Paper single-task 10k / 1×4090	24 GB	~9 h	cite / N as course default
Paper multi-task 30k / 2×4090 + full real	2×24 GB+	≫24 h + robot	N

Empiricist exam bite: "Quote **60.5 / 37.3 / +23.2**, Tab.3 **0 → 80**, occlusion **90**, USB multi-task **10**, ForceVLA-Data **244/~140k**; RC1+RC2+RC3 = π_0 bakeoff + fusion stage + insertion metrics with force_fusion/force_source logged; Ablation Bot AUTHORITATIVE ranks 1–7; refuse $\pi_{0.5}$ bake-off / FM ↔ FAST / stub=60.5% laundering."

6.5 Visionary — balanced (~12–14) — BEHAVIOR contact-rich

Seat charge: Decision-oriented **BEHAVIOR contact-rich / force-tactile** hooks. Late force MoE on modern VLA substrates. Pair with MoPA-PD (clutter transit) and DART (recovery). **No Discord/Drive checklist.**

BEHAVIOR connection (honest)

BEHAVIOR-class household activities are dense with **contact-rich** phases: plug/USB-like inserts, drawer/cabinet seating, wiping spills, peeling/food prep, press buttons. Vision-only VLAs (π_0 / $\pi_{0.5}$ line) inherit semantic open-world strength but still fail when **forces**, **occlusion**, and **compliance** dominate. ForceVLA’s durable idea:

Keep the modern VLA/FM controller; make calibrated 6-axis F/T a first-class late token with sparse expert fusion so contact phases can re-route computation.

BEHAVIOR need	ForceVLA-shaped response
Plug / peg / USB / charger insert	F/T-conditioned insert chunks; occlusion robustness
Drawers / dishwashers / lids	Contact detection + compliant push/pull via expert routing
Wiping / cleaning surfaces	Continuous force maintenance (wipe/peel analogs)
Crowded counters (visual clutter)	Lean on force when wrist cam occludes contact
Hybrid with $\pi_{0.5}$ / RTC stacks	TENTATIVE recipe: HL semantic loop + low-level FM expert augmented with FVLMoE when wrist F/T present — not claimed by paper
Hardware	Prefer arms with integrated or retrofit F/T; budget calibration; paper’s estimated-wrench caveat

Visionary bets (balanced; not monopoly)

1. **Late force adapter on $\pi_{0.5}$ / GR00T / OpenVLA-OFT:** Do **not** prepend force into VLM; add FVLMoE-style late MoE (E small, k=1) on action expert — Tab.3 moral.
2. **Mandatory wrench in contact-stage demos:** BEHAVIOR contact stages without FT/tactile → collect **estimated wrench or low-cost FT** subset.
3. **Metric card beyond SR:** peak $|F|$, insertion depth, slip events, peel-length analogs — Tab.1 honesty.
4. **Occlusion + force:** When vision drops (Tab.2 **90%**), force routing should dominate — log expert_id.
5. **Do not naive-concat onto FAST/compact tokenizers** without pretrain — paper π_0 -fast **31** → **14.2%**.
6. **Pair with MoPA-PD / DART siblings:** Force for **contact**; planner-distill for **clutter transit**; DART for **recovery** — three cards, not one slogan.
7. **Sim-first smoke:** OmniGibson / BEHAVIOR-sim contact with simulated wrench → then robot Partial.
8. **Empiricist honesty:** Course validates fusion stage + metrics + schema; leaderboard needs real contact sensors — never Fig.5 60.5% from stub wrench.
9. **USB-hard curriculum:** Force alone insufficient for fine alignment (multi-task USB **10%**) — curriculum / tactile / precision priors still needed.
10. **One narrative line:** VLAs see; ForceVLA **feels through a routed expert**; 2026 should **late-fuse wrench into the action expert**, measure contact, and keep VL pretraining intact.

Follow-up research

1. Port FVLMoE onto open π_0 /OpenPI with public F/T datasets.
2. Replace estimated wrench with high-fid transducers; ablate noise.
3. Add tactile image tokens beside F/T in the same MoE.
4. Contact-phase rewards / eval suites inside BEHAVIOR (insert, wipe, open).
5. Study Expert-0 generalist bias; load-balance for rare contact events.
6. Low-cost retrofit F/T on common arms (paper's stated direction).
7. Multi-task USB-hard curriculum — force alone insufficient for fine alignment.

New apps (beyond Discord/Drive)

- Kitchen plug / appliance insert assist.
- Wipe / scrub force profiles for cleaning robots.
- Food prep peel/cut force phases.
- Occluded cable seating behind furniture.
- Safe contact exploration under torque limits (Height Gen. lesson).

(No Discord / Drive checklist content.)

Visionary one-liner: For BEHAVIOR contact-rich homes, pair modern VLAs with wrist F/T + late MoE fusion so insertion/wipe/occlusion phases stop being pure vision failures — ForceVLA is the 2025 blueprint beside MoPA-PD (clutter transit) and DART (recovery), not a $\pi_{0.5}$ replacement.

Visionary exam bite: "ForceVLA = **late force-MoE recipe** for contact-rich beside **planner-distill** (MoPA-PD) and **recovery-demo** (DART); cite **60.5/37.3/+23.2**, Tab.3 **0** → **80**, occlusion **90**, USB **10**, ForceVLA-Data **244/~140k**; Empiricist smokes directional only; **no $\pi_{0.5}$ bake-off**; **no Discord/Drive**."

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export FORCEVLA_WORK_ROOT=$SCRATCH/forcevla-smoke
cd /workspace/forcevla-quiet
bash env_outline/setup_mamba.sh
```

```
# mamba activate forcevla-scout
mkdir -p logs runs checkpoints demos
# EDIT: pin ForceVLA / OpenPI  $\pi_0$  when public; prefer force stub if data unreleased
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	mamba sketch (Torch + OpenPI/ π_0 deps + force IO stubs)
env_outline/setup_mamba.sh	env create; OpenPI / ForceVLA clone notes
env_outline/run_checklist.md	GPU / TIME honesty preflight
env_outline/README.md	cluster usage
env_outline/slurm_data_ingest_smoke.sh	RC0
env_outline/slurm_fvlmoe_vs_pi0.sh	RC1 / #4 PRIMARY
env_outline/slurm_fusion_stage_ablation.sh	RC2 / #1 PRIMARY
env_outline/slurm_insertion_metrics.sh	RC3 PRIMARY
env_outline/slurm_data_scale_smoke.sh	RC4
env_outline/slurm_gen_occlusion.sh	RC5 / #5 Partial
env_outline/slurm_moe_router_log.sh	RC6 Partial
env_outline/slurm_multitask_joint_stub.sh	RC7 / #6 Partial

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
FVLMoE $\gg$ vision-only / $\geq$ naive F	RC1: directional SR or contact metrics; log force_fusion	Fig.5 60.5% absolute / 5-task
Late fusion $\gg$ early MoE	RC2: late $\geq$ early; early_moe does not win	Tab.3 80% absolute; full early-MoE train
Contact-rich metrics honesty	RC3: SR + wrench + peel fields filled	Paper n=20x5 Flexiv protocol
ForceVLA-Data schema	RC0: dim-6 wrench sync; traj count logged	Full 244-traj public release
Occlusion robustness	RC5 Partial: force-on $\geq$ force-off under mask	Tab.2 90% absolute
Multi-task joint	RC7 cite / stub	Tab.5 67.5% absolute
Full paper protocol	Document N	Never course

Pass: documented smoke with Ablation Bot **#1 + #4 + insertion metrics** confirmed directionally + force_fusion/force_source logged.

Fail / overclaim: “reproduced Fig.5 60.5%” / “beats $\pi_{0.5}$ ” / “ForceVLA is FAST” / stub wrench as ForceVLA-Data from Partial smoke.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Archaeologist	balanced · ~12-14	CLEAR/TENTATIVE; π_0 parent; no $\pi_{0.5}$ bake-off; FM $\neq$ FAST; MoPA-PD contrast; genealogy; exam bite
Stakeholder	balanced · ~12-14	Buy late MoE pattern; risks (FT cost, USB 10%, estimated wrench); decision ask

Role	Weight / timing guide	Focus
Scientific Reviewer	balanced · ~12-14	Tab.3+Fig.5+occlusion; pump/USB caveats; verdict Accept Alt contact-rich; fence $\pi_{0.5}$ / FM $\leftrightarrow$ FAST
Empiricist	balanced · ~12-14	H-fusion/avg/plug; RC1+RC2+RC3; Ablation Bot AUTHORITATIVE; GPU honesty; must-cite tattoo
Visionary	balanced · ~12-14	BEHAVIOR contact-rich; late force-MoE + MoPA-PD + DART; no Discord/Drive

Spine (paste into Slide Maker) — 5 bullets: 1. **Late 6-axis F/T + FVLMoE (4 experts, top-1) augments π_0 FM head for contact-rich**; fusion quality $\gg$ naive concat; **NOT** a FAST switch; contrast **MoPA-PD** (train privilege) vs **deploy-side force**.
2. **Must-cite:** avg **60.5** vs **37.3 / 40.2**; Tab.3 early MoE **0%** $\rightarrow$ **ForceVLA 80%**; occlusion **90%**; USB multi-task **10%**; ForceVLA-Data **244 traj / ~140k**; **no $\pi_{0.5}$ bake-off**.
3. **Taxonomy:** π_0 FM parent | π_0 -fast baseline (naive F hurts) | ForceVLA late MoE | MoPA-PD (**CLEAR contrast**) | $\pi_{0.5}$ (**related, unmeasured**) | JEPA/WM (**contrast poles only**).
4. **Lineage (Club Pack — all seats empty, balanced):** CLEAR parent π_0 ; inventum = late force token + FVLMoE additive FM guidance; **NOT** FAST head swap; **NOT** MoPA-PD / JEPA child.
5. **BEHAVIOR’26 + honesty:** ForceVLA = **late force-MoE recipe** for contact-rich beside MoPA-PD clutter transit + DART recovery; Empiricist RC1–RC3 directional; **do not invent $\pi_{0.5}$ bake-off** / Fig.5 60.5% from stub — **no Discord/Drive**.

GitHub title sketch: [ECE 605] - ForceVLA Making Club Pack

Full Slide Maker brief: /workspace/forcevla-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/forcevla-quiet/KANTA_PACK.md
PDF	/workspace/forcevla-quiet/KANTA_PACK.pdf
Slide brief	/workspace/forcevla-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/forcevla-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/forcevla-2505.22159-summary.md
Ablations	AUTHORITATIVE — /workspace/papers/forcevla-2505.22159-ablations.md
Paper notes	/workspace/forcevla-quiet/paper_notes.md
Env outlines	/workspace/forcevla-quiet/env_outline/
Paper PDF	/workspace/papers/forcevla-2505.22159.pdf
Text extract	/workspace/papers/forcevla-2505.22159.txt · paper_clean.txt
Page figs	/workspace/papers/forcevla-figs/page-01.png ... page-20.png
Sibling links	/workspace/pi0-quiet/ · /workspace/pi05-quiet/ · /workspace/fast-quiet/ · /workspace/distill-mopa-quiet/ · /workspace/dart-quiet/ · /workspace/openvla-quiet/ · /workspace/openvla-ofc-quiet/
Project / code	https://sites.google.com/view/forcevla2025 · https://github.com/ft-robotic/ForceVLA

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 — Undated LAST Alternative approaches · ALL seats EMPTY · balanced five-role Club Pack · Ablation Bot AUTHORITATIVE ranks 1–7 · CLEAR parent π_0 · NOT FAST switch · no $\pi_{0.5}$ bake-off · CLEAR contrast MoPA-PD · must-cite 60.5 vs 37.3/40.2 · Tab.3 0 → 80 · occlusion 90 · USB multi-task 10 · ForceVLA-Data 244/~140k · Visionary BEHAVIOR contact-rich · no Discord/Drive · outlines only — no cluster execution.